

- (6 pts) Sketch (below) $w(z)$ for a Gaussian beam from $z = -3z_r$ to $+3z_r$. Indicate the radius of curvature of the wavefront at z_r , and the correct behavior of the beam in the far field region. Identify the $z = 0$ position.
- (4 pts) Write the equation for the TEM_{00} Gaussian beam intensity, $I(r)$. Sketch $I(r)$ for the cross section of a TEM_{00} Gaussian beam and indicate I_0 and ω on the plot. (Label axes!)
- (4 pts) If the minimum beam waist of a HeNe laser (632.8nm) TEM_{00} beam is 0.5 mm at the output mirror, what will be the diameter of the beam at a distance of 10 m (assume $n=1$)?
- (6 pts) Explain (feel free to use sketches, plots, and words) how you proved that the pinhole trace of the HeNe laser beam TEM_{00} mode was Gaussian, ie, followed the equation of Problem 2 above.